

AI-Based Multi-Agent Diabetic Retinopathy Diagnostic and Progression Analysis System

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Chapter 1

Introduction

Diabetic Retinopathy (DR) is one of the leading causes of blindness worldwide. Early detection and continuous monitoring are critical to prevent irreversible vision loss. Traditional diagnosis depends heavily on ophthalmologists and manual retinal examination, making the process time-consuming and inaccessible in many regions.

This project proposes an **AI-Based Multi-Agent Diabetic Retinopathy Diagnostic and Progression Analysis System** that performs automated DR detection, severity grading, explainable AI visualization, progression tracking, disease progression simulation, predictive treatment recommendation, and intelligent report interpretation using Large Language Models (LLMs).

The system focuses primarily on retinal image analysis using deep learning and explainable AI while integrating modern AI concepts such as multi-agent systems, progression forecasting, and natural language medical assistance.

Key Features:

- Automated Diabetic Retinopathy detection
- Severity grading using CNN-based architecture
- Explainable AI using Grad-CAM heatmaps
- Progression tracking across multiple scans
- Disease spread simulation
- LLM-powered medical report interpretation
- Predictive treatment recommendation system
- Doctor dashboard with patient history
- PDF report generation
- Future-ready scalable architecture

Chapter 2

Problem Statement

Problem:

Current DR diagnosis relies heavily on ophthalmologists and manual retinal analysis. Tracking disease progression manually over long periods becomes difficult and error-prone. Existing systems lack explainability, intelligent assistance, longitudinal monitoring, and predictive capabilities.

Solution:

Develop an AI-powered multi-agent diagnostic system capable of:

- Detecting DR automatically from retinal images
- Classifying severity stages
- Explaining predictions using heatmaps
- Tracking progression over time
- Simulating future retinal damage spread
- Generating intelligent treatment suggestions
- Using LLMs to explain medical findings in simple language
- Managing patient history through a doctor dashboard

Chapter 3

Objectives

1. Develop a CNN-based DR classification model.
2. Perform severity grading of DR stages.
3. Integrate Grad-CAM for explainable AI visualization.
4. Build progression tracking using historical scan data.
5. Simulate retinal disease spread over time.
6. Generate predictive treatment recommendations.
7. Integrate LLM for report explanation and patient interaction.
8. Build a doctor dashboard for patient monitoring.
9. Generate downloadable reports.

Chapter 4

System Features

Feature	Description
CNN-Based Diagnosis	Detects DR from retinal images using EfficientNetB0.
Severity Grading	Predicts DR severity levels (0-4).
Explainable AI	Uses Grad-CAM heatmaps to highlight affected retinal regions.
Progression Tracking	Tracks disease evolution using historical patient scans.
Disease Spread Simulation	Simulates possible future retinal damage progression visually.
LLM Integration	Explains reports and predictions in simple language for doctors/patients.
Predictive Treatment Recommendation	Suggests medical follow-up based on severity and progression trends.
Doctor Dashboard	Allows doctors to monitor patient history and reports.
PDF Report Generation	Generates downloadable medical reports with predictions and heatmaps.
Image Quality Validation	Detects blurry or invalid retinal images before prediction.
Multi-Agent Architecture	Modular agents handle preprocessing, diagnosis, explainability, tracking, simulation, and reporting.

Chapter 5

System Architecture and Workflow

5.1 Architecture Overview

The system follows a modular multi-agent architecture where each agent performs a dedicated task.

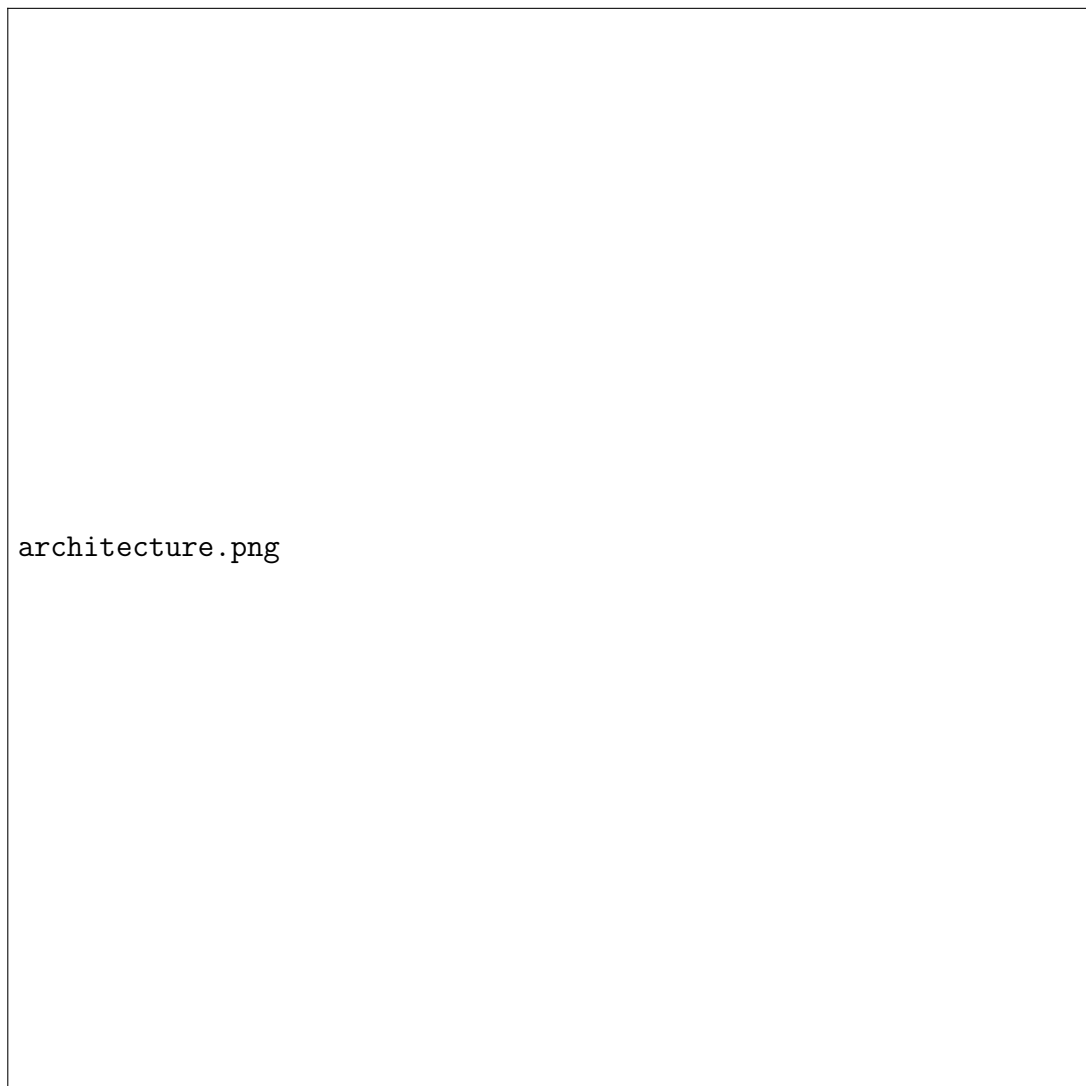


Figure 5.1: System Architecture

5.2 Workflow

1. User uploads retinal image
2. Preprocessing Agent:
 - Resize image
 - Normalize image
 - Contrast enhancement
3. Quality Check Agent:
 - Blur detection
 - Resolution validation
4. Diagnosis Agent:
 - EfficientNetB0 performs DR classification
5. Severity Agent:
 - Predicts DR stage
6. Explainability Agent:
 - Generates Grad-CAM heatmap
7. Progression Tracking Agent:
 - Compares current scan with historical scans
8. Disease Spread Simulation Agent:
 - Predicts possible retinal damage expansion
9. Predictive Treatment Agent:
 - Suggests medical follow-up recommendations
10. LLM Agent:
 - Converts technical findings into understandable explanations
11. Report Generation Agent:
 - Creates PDF report
12. Dashboard Agent:
 - Displays patient history and progression

Chapter 6

Dataset

6.1 Primary Dataset

APTOS 2019 Blindness Detection Dataset

Used for:

- DR classification
- Severity grading

6.2 Classes

Label	Severity
0	No DR
1	Mild
2	Moderate
3	Severe
4	Proliferative DR

Chapter 7

Implementation Details

7.1 Technology Stack

- Python
- TensorFlow / Keras
- OpenCV
- Flask
- SQLite
- Matplotlib
- ReportLab
- HuggingFace/OpenAI API (optional LLM integration)

7.2 Implementation Phases

7.2.1 Phase 1: Dataset Preparation

- Download APTOS dataset
- Organize train/validation/test folders
- Create CSV label mapping

Folder Structure:

```
dataset/  
  train/  
  val/  
  test/  
  labels.csv
```

7.2.2 Phase 2: Preprocessing Pipeline

- Resize images to 224x224
- Normalize pixel values
- Apply augmentation

Example augmentations:

- Rotation
- Horizontal flip
- Zoom

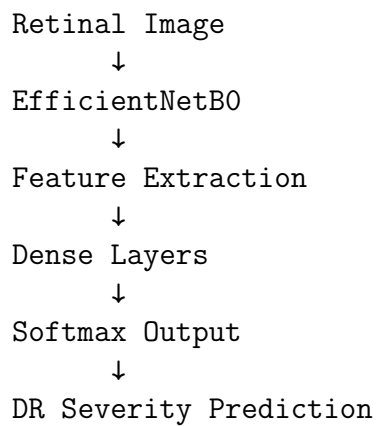
7.2.3 Phase 3: CNN Model Implementation

Model Used: EfficientNetB0

Why EfficientNetB0?

- High accuracy
- Lightweight
- Good transfer learning performance
- Suitable for medical imaging

Workflow



7.2.4 Phase 4: Model Training

- Use transfer learning
- Freeze initial layers
- Fine-tune final layers
- Train for 15-25 epochs

Target Accuracy:

- 80%-90%

7.2.5 Phase 5: Grad-CAM Integration

Purpose:

Highlight retinal regions responsible for prediction.

Workflow

Prediction

↓

Gradient Calculation

↓

Feature Importance

↓

Heatmap Overlay

Output:

- Red regions indicate affected areas

7.2.6 Phase 6: Progression Tracking

Goal:

Monitor disease changes over time.

Implementation

- Store all scan history in database
- Compare previous severity with current severity
- Plot progression graph

Example:

Jan → Mild

Mar → Moderate

Jun → Severe

7.2.7 Phase 7: Disease Spread Simulation

Goal:

Simulate possible future retinal damage.

Basic Implementation

- Use progression history
- Use image masks and affected regions
- Predict possible lesion spread visually

Methodology

- Use Grad-CAM outputs
- Expand affected regions probabilistically
- Generate future retinal state visualization

7.2.8 Phase 8: Predictive Treatment Recommendation

Goal:

Provide basic follow-up suggestions.

Implementation

Rule-based system:

Severity	Recommendation
0	Routine yearly screening
1	Monitor blood sugar and follow-up
2	Ophthalmologist consultation advised
3	Immediate specialist consultation
4	Urgent retinal treatment required

7.2.9 Phase 9: LLM Integration

Goal:

Convert technical predictions into human-readable explanations.

Example

Input:

Severity: Moderate DR

Confidence: 91%

LLM Output:

The retinal scan shows moderate diabetic retinopathy with visible abnormalities in blood vessels.
A consultation with an ophthalmologist is recommended.

Possible LLM Usage

- Report summarization
- Patient-friendly explanations
- Doctor notes generation
- Chat assistant for DR awareness

7.2.10 Phase 10: Flask Web Application

Features:

- Image upload
- Prediction display
- Heatmap visualization
- Progression graph
- Report download

7.2.11 Phase 11: Doctor Dashboard

Features:

- View patient history
- Compare scans
- Monitor progression
- Review reports

7.2.12 Phase 12: Report Generation

Generated PDF includes:

- Patient details
- Prediction
- Confidence score
- Heatmap
- Progression graph
- Treatment recommendation
- LLM-generated explanation

Chapter 8

Database Design

8.1 Patients Table

Field	Type
patient_id	Integer
name	Text
age	Integer
gender	Text
diabetes_history	Text

8.2 Reports Table

Field	Type
report_id	Integer
patient_id	Integer
image_path	Text
severity	Integer
confidence	Float
heatmap_path	Text
recommendation	Text
llm_summary	Text
date	Date

8.3 Progression Table

Field	Type
progression_id	Integer
patient_id	Integer
previous_severity	Integer
current_severity	Integer
status	Text
date	Date

Chapter 9

Folder Structure

```
project/  
  
  dataset/  
  models/  
  static/  
    uploads/  
    heatmaps/  
    reports/  
  
  templates/  
    index.html  
    dashboard.html  
    result.html  
  
  app.py  
  train.py  
  gradcam.py  
  llm_agent.py  
  progression.py  
  simulation.py  
  report_generator.py  
  database.py  
  requirements.txt
```

Chapter 10

Future Enhancements

- Multi-disease detection using ODIR and RFMID
- OCT scan integration
- GNN-based patient relationship modeling
- Real-time webcam analysis
- Blockchain-secured reports
- Multi-hospital cloud deployment
- Federated learning

Chapter 11

Expected Outcomes

- Accurate DR classification
- Explainable heatmap visualization
- Intelligent report explanation
- Disease progression monitoring
- Future damage simulation
- Automated treatment guidance
- Complete doctor dashboard

Chapter 12

Conclusion

The proposed AI-Based Multi-Agent Diabetic Retinopathy Diagnostic and Progression Analysis System provides an intelligent and scalable approach for retinal disease analysis. By combining CNN-based diagnosis, explainable AI, progression tracking, disease spread simulation, predictive treatment recommendation, and LLM integration, the system moves beyond traditional image classification toward a comprehensive ophthalmic decision-support platform.